Computational Statistics

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2026-04-24

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Preface

This is a concise course about statistical inference, which was developed for the course STAT 442/851 at University of Saskatchewan.

Key Features

- **Traditional Mathematical Rigor:** Emphasis is placed on the rigorous mathematical derivation and proof of the fundamental theorems underpinning statistical inference. For example, students will engage deeply with the analytic proofs of the Neyman-Pearson Lemma, the asymptotic normality of Maximum Likelihood Estimators (MLE), and the formulation of the Cramér-Rao Lower Bound.
- **Integration of Computational Tools:** Utilization of computational simulations and graphical representations to elucidate and validate complex statistical concepts. For instance, the course employs simulation studies to empirically demonstrate the advantages of shrinkage estimators and to visualize the asymptotic distributions governed by Maximum Likelihood Estimation (MLE) theory and Wilks' Theorem. Furthermore, discussions will highlight the operational relevance of classical mathematical theorems within modern computational frameworks, such as the implications of Bartlett Identities in deep learning algorithms.
- **Priority of Topics in Light of Modern Practice in Statistics and Machine Learning:** The curriculum is strategically curated to emphasize methodologies that offer broad generalizability and utility in contemporary applications. Prominence is given to versatile frameworks such as Bayesian inference, regularization techniques, Maximum Likelihood Estimation (MLE), likelihood-based hypothesis testing, information criteria, and rigorous model assessment. Concurrently, essential classical foundations, including the Neyman-Pearson Lemma and the theory of Uniformly Minimum Variance Unbiased Estimators (UMVUE), are deliberately retained to ensure a comprehensive theoretical grounding.

Audience

This course requires a strong command of multivariate calculus, alongside a rigorous foundation in intermediate probability theory including asymptotic theorey for probability. Students should also possess prior exposure to applied statistical methods and familiar with basic statistical concepts such as p-value and confidence internal.

About the author

Longhai Li is a professor at the University of Saskatchewan in Canada. He received his Ph.D. degree in statistics from the University of Toronto. His research activities focus on developing and applying statistical machine-learning methods for bioinformatics and epidemiology applications, with particular interests on statistical learning, cross-validation, hierarchical modelling, survival modelling, model checking, residual diagnostics, model comparison, zero-inflated models, high-throughput data, microbiome data. His research has been funded by NSERC, CANSSI, CFI, CFREF, and MITACS. His research papers have appeared in highly reputed journals, such as Journal of American Statistical Association, Bayesian Analysis, Statistics in Medicine, Statistics and Computing, American Statistician, Journal of Applied Statistics, Scientific Reports, and BMC Bioinformatics.

Notation and Symbols

Throughout this document, we use **boldface lowercase** letters (e.g., **x**) to denote vectors and **boldface uppercase** letters (e.g., **X**, **I**) to denote matrices. Scalar variables and parameters are written in standard italics (e.g., x, θ).

Symbol	Description
Variables & Parameters	
x, X	Scalar random variable or observation
x, **X**	Vector random variable or observation column vector
θ	Scalar unknown parameter
θ	Vector of unknown parameters, $\theta \in \mathbb{R}^p$
Θ	Parameter space
0	Zero vector or zero matrix (dimension implied by context)
$\mathbf{I}_p$	Identity matrix of size $p \times p$
Functions & Operators	
$f(x\|\theta)$	Probability density function (pdf) or probability mass function (pmf)
$\ell(\theta)$	Log-likelihood function, $\ell(\theta) = \log f(\mathbf{x}\|\theta)$
$E^{\mathbf{X}\|\theta}[\cdot]$ or $E_\theta[\cdot]$	Expectation w.r.t. **X** conditional on θ
$\mathrm{Var}^{\mathbf{X}\|\theta}(\cdot)$ or $\mathrm{Var}_\theta(\cdot)$	Variance (or Covariance Matrix) w.r.t. **X** conditional on θ
$\mathrm{Cov}_\theta(\cdot,\cdot)$	Covariance between two random variables or vectors
$\mathbf{A}^\top$ or $\mathbf{A}^T$	Transpose of vector or matrix **A**
$\mathrm{tr}(\mathbf{A})$	Trace of matrix **A** (sum of diagonal elements)
$\mathbf{A} \succeq \mathbf{B}$	Matrix inequality; $\mathbf{A} - \mathbf{B}$ is positive semi-definite
Calculus & Gradients	
$\nabla_\theta f$	Gradient vector with respect to θ
$\nabla^2_\theta f$	Hessian matrix (second derivatives) with respect to θ
$\frac{\partial}{\partial\theta}$	Partial derivative operator
$\mathbf{J}_f(\mathbf{x})$	Jacobian matrix of a vector-valued function f
$\nabla \cdot \mathbf{g}$	Divergence of a vector field **g**, $\sum \frac{\partial g_i}{\partial x_i}$
Statistical Quantities	
$\mathbf{U}(\theta)$	Score vector, $\nabla_\theta \ell(\theta)$

Symbol	Description
$\mathbf{J}(\theta)$	Observed Information matrix, $-\nabla_\theta^2 \ell(\theta)$
$\mathcal{I}(\theta)$	Fisher Information matrix, $E[\mathbf{U}\mathbf{U}^\top] = E[\mathbf{J}]$
$T(\mathbf{X})$	Estimator or statistic
$m(\theta)$	Expectation of an estimator, $E[T(\mathbf{X})]$
$\mathbf{D}(\theta)$	Jacobian of the expectation vector $m(\theta)$

1 Introduction to Statistical Inference

1.1 Population Model (Data Model)

We begin with observations (units) $X_1, X_2, \ldots, X_n$. These may be vectors. We regard these observations as a realization of random variables.

Definition 1.1 (Population Distribution)**.** We assume that $X_1, X_2, \ldots, X_n \sim f(x)$. The function $f(x)$ is called the **population distribution**.

Assumptions and Scope

For simplicity, we often assume the data are Independent and Identically Distributed (i.i.d.). The assumption of identical distribution can be relaxed to regression settings in which the distributions of x_i's are independent but dependent on covariate x_i.

In **Parametric Statistics**, we assume $f(x)$ is of a known analytic form but involves unknown parameters.

Example 1.1 (Parametric Model: Normal)**.** Consider the Normal distribution:

$$f(x;\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{1.1}$$

Here, the parameter space is $\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma \in [0, +\infty)\}$. The goal is to learn aspects of the unknown θ from observations $X_1, \ldots, X_n$.

Example 1.2 (Parametric Model: Bernoulli)**.** Consider a sequence of binary outcomes (e.g., Success/Failure) where each $X_i \in \{0, 1\}$. We assume $X_i \sim \text{Bernoulli}(\theta)$. The probability mass function is:

$$f(x;\theta) = \theta^x (1-\theta)^{1-x} \tag{1.2}$$

Here, the parameter space is $\Theta = [0, 1]$, where θ represents the probability of success.

1.2 Probabilistic Model vs. Statistical Inference

There is a fundamental distinction between probability and statistics regarding the parameter θ. We can visualize this using a "shooting target" analogy:

- θ **(The Center):** The true, unknown bullseye location.

- x **(The Shots):** The observed holes on the target board.
- **Probability (Deductive):** The center θ is **known**. We predict where the shots x will land.
- **Statistics (Inductive):** The shots x are **observed** on the board. The center θ is unknown. We hypothesize different potential centers to see which one best explains the shots.

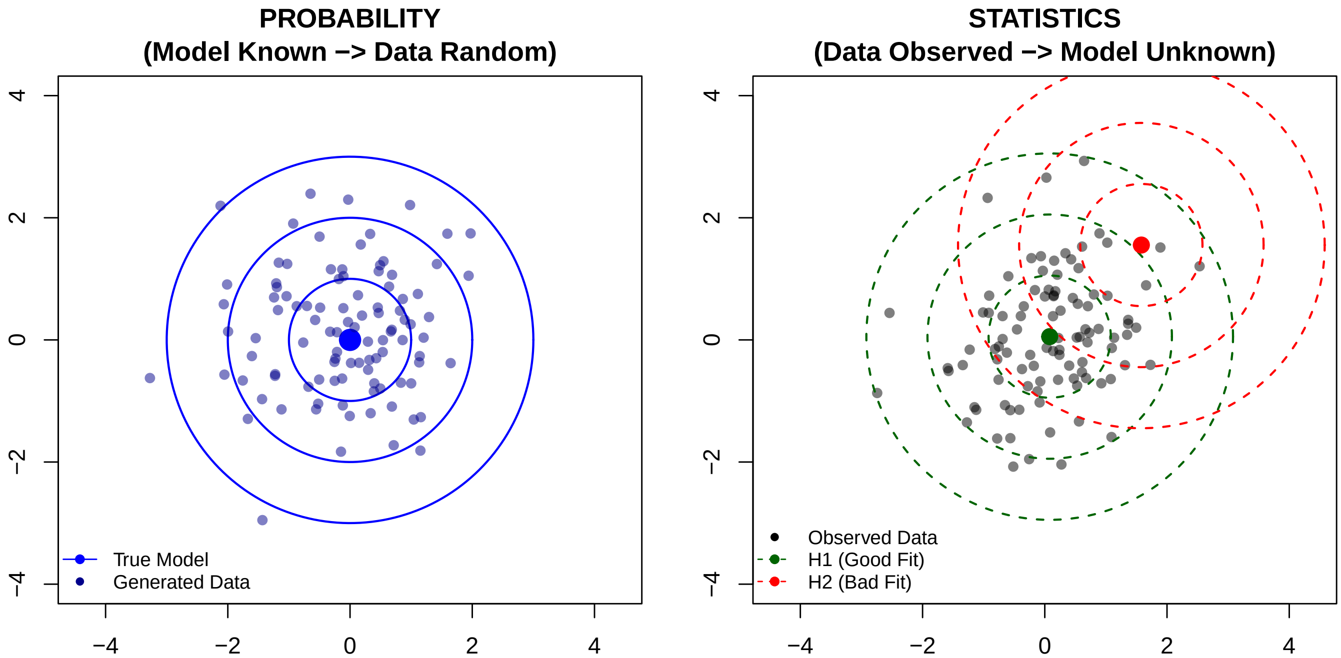

Figure 1.1: Probability vs Statistics. Left: Probability—The model is fixed (Blue center/contours), generating random data. Right: Statistics—Data is fixed (Black points); we test two hypothesized models: H1 (Green) centered at the sample mean (Good Fit) and H2 (Red) shifted by (1.5, 1.5) (Bad Fit).

1.3 A Motivating Example: The Lady Tasting Tea

To illustrate the concepts of statistical inference, we consider the famous experiment described by R.A. Fisher.

A lady claims she can distinguish whether milk was poured into the cup before or after the tea. To test this claim, we prepare n cups of tea.

- **Random Variable:** Let $X_i = 1$ if she identifies the cup correctly, and 0 otherwise.
- **Parameter:** Let θ be the probability that she correctly identifies a cup.
- **The Data:** Suppose we observe that she identifies **70%** of cups correctly ($\bar{x} = 0.7$), which is a summary of the observed vector of x_i, for example,

$$x = (0, 1, 1, 0, 1, 1, 0, 1, 1, 1) \tag{1.3}$$

1.3.1 Small Sample (n=10)

We observe **7 out of 10** correct ($k = 7$).

$$\bar{x} = 0.7 \tag{1.4}$$

1.3.2 Large Sample (n=40)

We observe **28 out of 40** correct ($k = 28$).

$$\bar{x} = 0.7 \tag{1.5}$$

1.4 Questions to Answer in Statistical Inference

Using this example, we identify the four main types of statistical inference.

Point Estimation

We want to use a single number to capture the parameter: $\hat{\theta} = \theta(X_1, \dots, X_n)$.

- *Tea Example:* Our best guess for her success rate is $\hat{\theta} = 0.7$.

Hypothesis Testing

We want to test a theory about the parameter: H_0 vs H_1.

- *Tea Example:* Is she just guessing? We test $H_0 : \theta = 0.5$ vs $H_1 : \theta > 0.5$.

Model Assessment

We want to test a theory about the parameter: H_0 vs H_1.

- *Example:* Can we use a reduced model? What level of complexity of $f(x; \theta)$ is necessary?

Interval Estimation

We want to construct an interval likely to contain the parameter: $\theta \in (L, U)$.

- *Tea Example:* We might say her true skill θ is likely between 0.45 and 0.95.

Prediction

We want to predict a new observation Y_{n+1} given previous data.

- *Tea Example:* If we give her an $(n + 1)$-th cup, what is the probability she identifies it correctly?

1.5 The Likelihood Function

The bridge between probability and statistics is the Likelihood Function.

Definition 1.2 (Likelihood Function)**.** Let $f(x_1, \dots, x_n; \theta)$ be the joint probability density (or mass) function of the data given the parameter θ. When we view this function as a function of θ for fixed observed data $x_1, \dots, x_n$, we call it the **likelihood function**, denoted $L(\theta)$.

$$L(\theta) = f(x_1, \dots, x_n; \theta) \tag{1.6}$$

Example: Lady Tasting Tea

For our Tea Tasting data, the likelihood is proportional to the Binomial probability:

$$L(\theta) = \binom{n}{k} \theta^k (1-\theta)^{n-k} \tag{1.7}$$

1.5.1 n=10 (k=7)

Here, $L(\theta) = \binom{10}{7}\theta^7(1-\theta)^3$.

θ	Calculation $\binom{10}{7}\theta^7(1-\theta)^3$	$L(\theta)$
0.0	$120 \times 0^7 \times 1^3$	0.0000
0.2	$120 \times 0.2^7 \times 0.8^3$	0.0008
0.4	$120 \times 0.4^7 \times 0.6^3$	0.0425
0.6	$120 \times 0.6^7 \times 0.4^3$	0.2150
0.7	$120 \times 0.7^7 \times 0.3^3$	**0.2668** (Max)
0.8	$120 \times 0.8^7 \times 0.2^3$	0.2013
1.0	$120 \times 1^7 \times 0^3$	0.0000

1.5.2 n=40 (k=28)

Here, $L(\theta) = \binom{40}{28}\theta^{28}(1-\theta)^{12}$. Notice how the likelihood becomes **narrower** (more peaked) with more data, even though the peak remains at 0.7.

θ	Calculation $\binom{40}{28}\theta^{28}(1-\theta)^{12}$	$L(\theta)$
0.0	$5.5868535 \times 10^9 \times 0^{28} \times 1^{12}$	0.0000
0.2	$5.5868535 \times 10^9 \times 0.2^{28} \times 0.8^{12}$	0.0000
0.4	$5.5868535 \times 10^9 \times 0.4^{28} \times 0.6^{12}$	0.0001
0.6	$5.5868535 \times 10^9 \times 0.6^{28} \times 0.4^{12}$	0.0576

θ	Calculation $\binom{40}{28}\theta^{28}(1-\theta)^{12}$	$L(\theta)$
0.7	5.5868535×10^9 $\times 0.7^{28} \times 0.3^{12}$	**0.1366** (Max)
0.8	5.5868535×10^9 $\times 0.8^{28} \times 0.2^{12}$	0.0443
1.0	5.5868535×10^9 $\times 1^{28} \times 0^{12}$	0.0000

Questions

- Is an estimator like $\bar{x}$, which is called Maximum Likelihood Estimator (MLE), a good estimator in general?
- What do you discover from actually observing the two likelihood unctions of different sample size n?
- Is the likelihood function central to all inference problems?
- What are the essential 'parameters' of the likelihood function?

There are two primary frameworks for "How" to perform these inferences.

1.6 Frequentist Inference

- **Concept:** θ is unknown but fixed; Data X is random.
- **Sampling Distribution:** We analyze how $\widehat{\theta}$ behaves under hypothetical repeated sampling.

Example: Frequentist Test of Lady Tasting Tea

We test $H_0 : \theta = 0.5$ (Guessing) vs $H_1 : \theta > 0.5$ (Skill). We analyze the behavior of $\bar{X}$ assuming H_0 is true. The rejection region (one-sided) is shaded red.

1.6.1 n=10 (k=7)

We calculate the P-value: Probability of observing ≥ 7 correct out of 10, assuming $\theta = 0.5$.

1.6.2 n=40 (k=28)

We calculate the P-value: Probability of observing ≥ 28 correct out of 40. With a larger sample size, the same proportion (0.7) provides **stronger evidence** against the null.

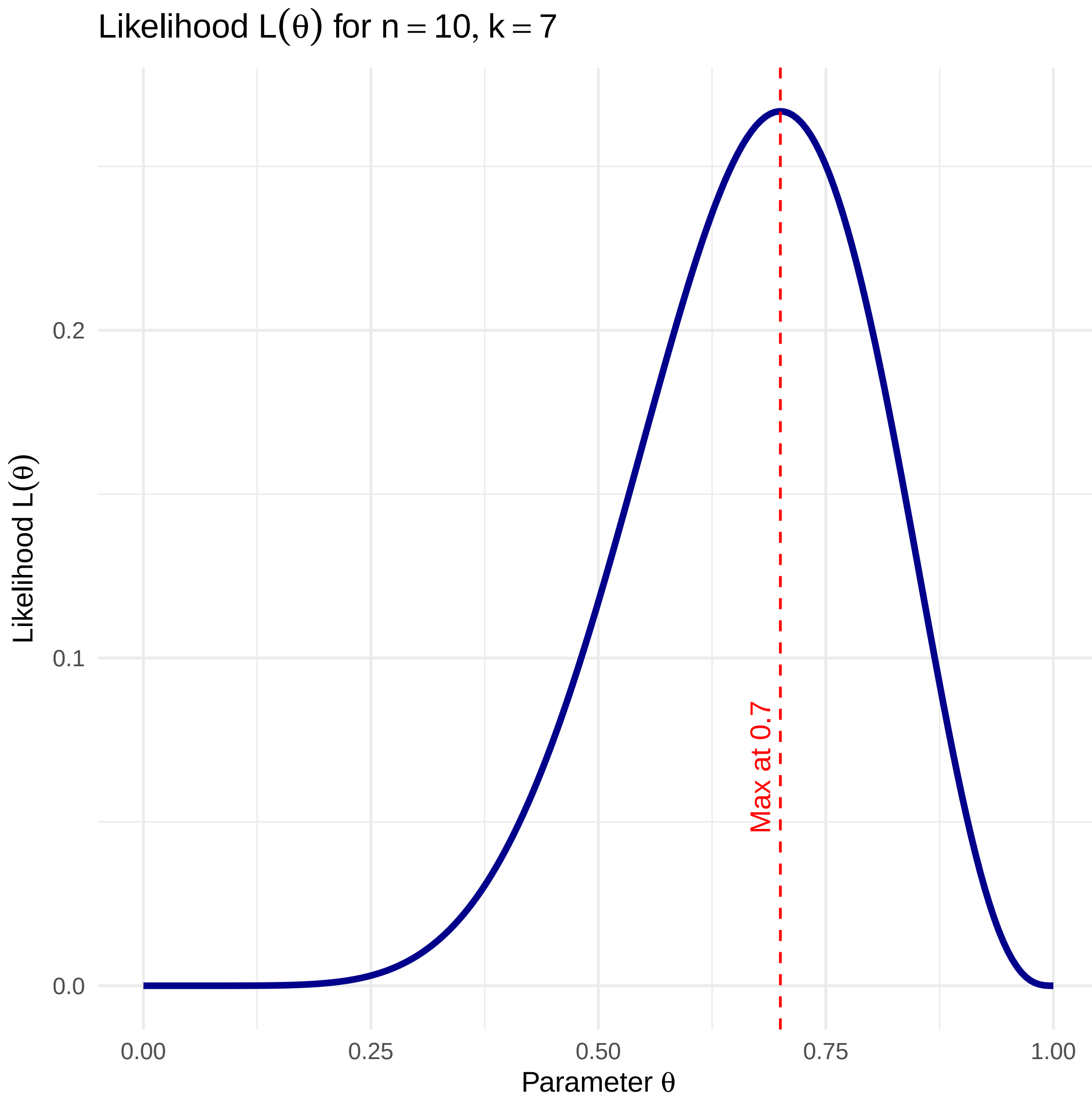

Figure 1.2: Likelihood Function (n= 10)

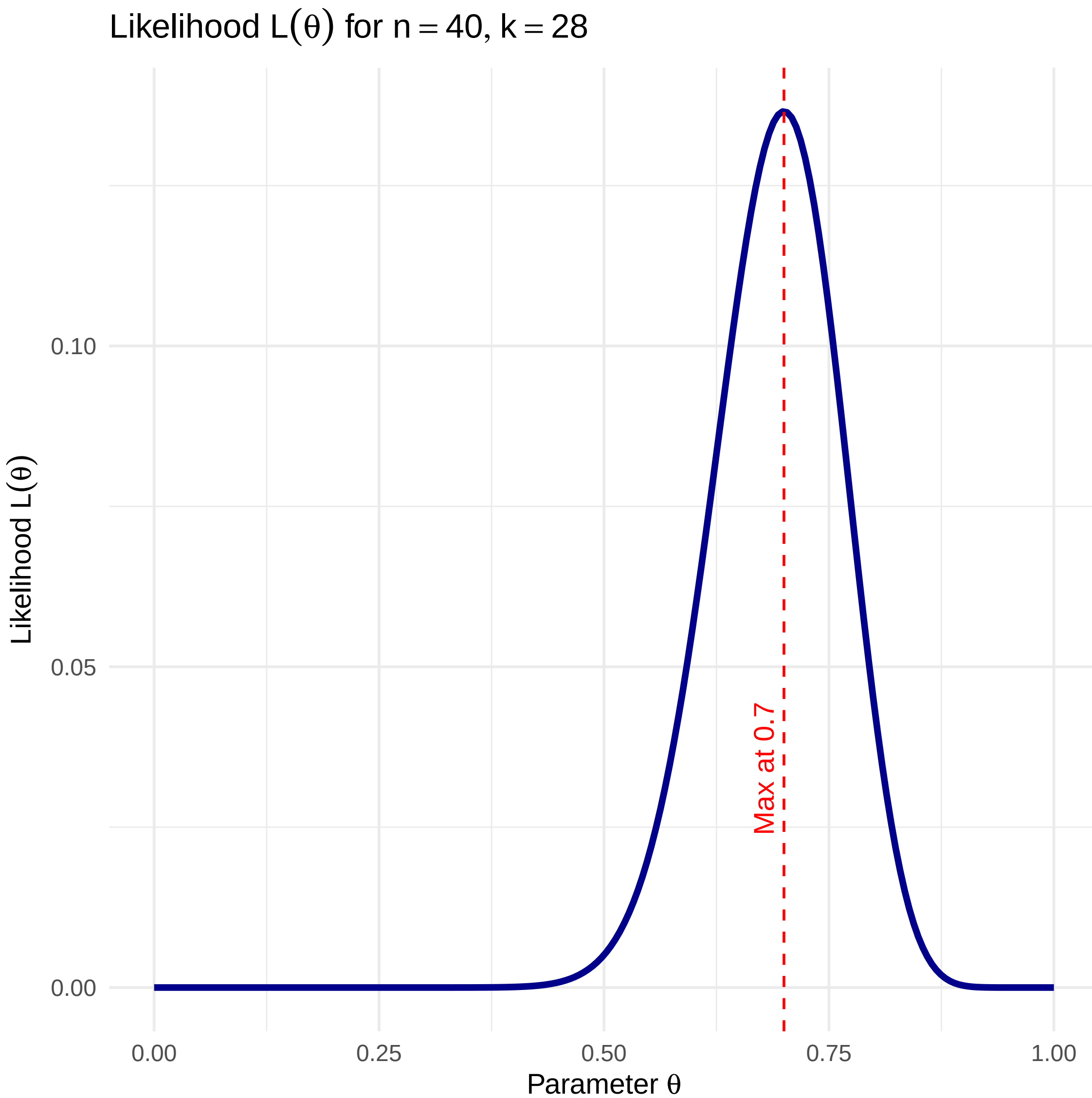

Figure 1.3: Likelihood Function (n= 40)

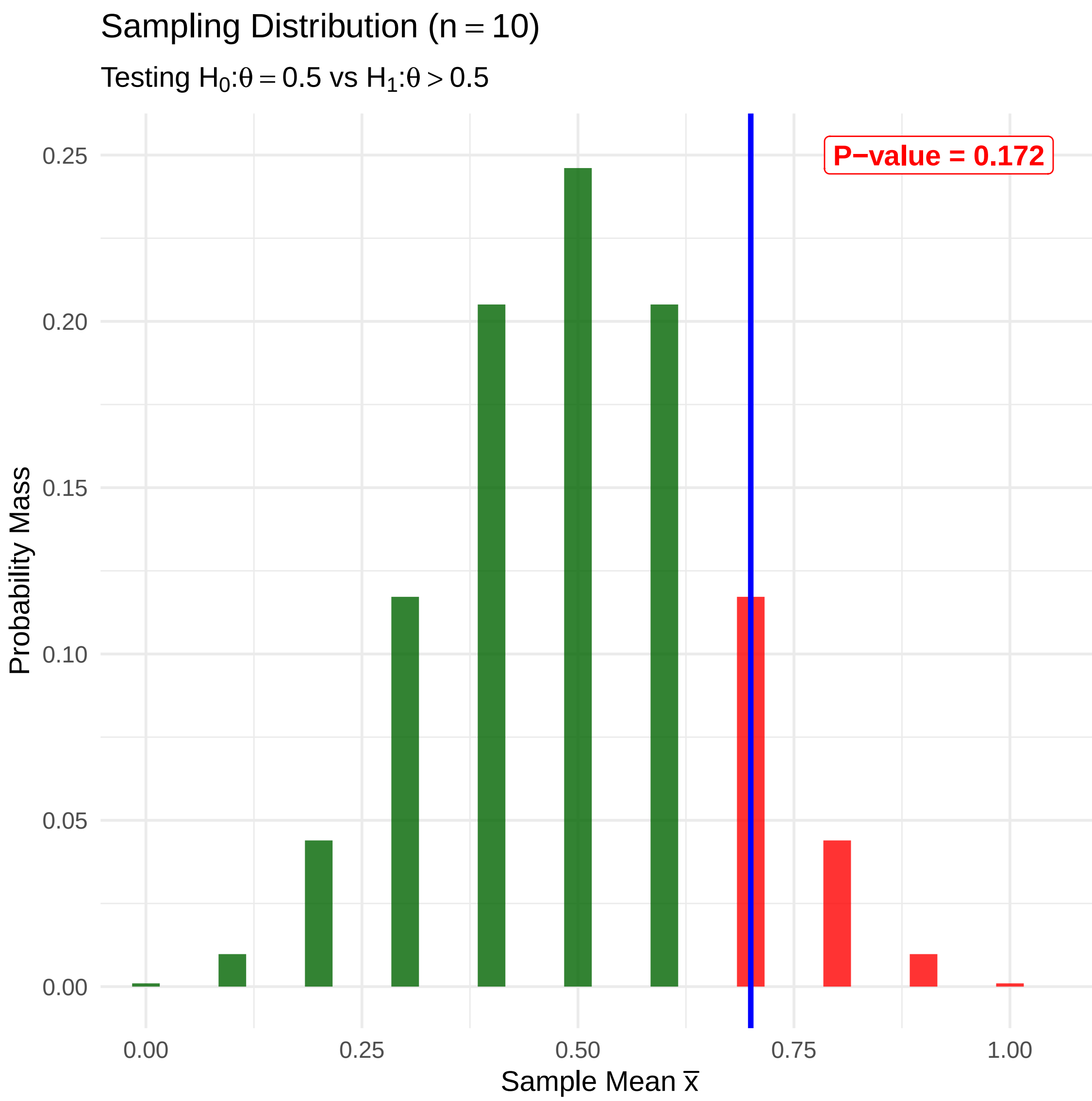

Figure 1.4: Sampling Distribution (n= 10)

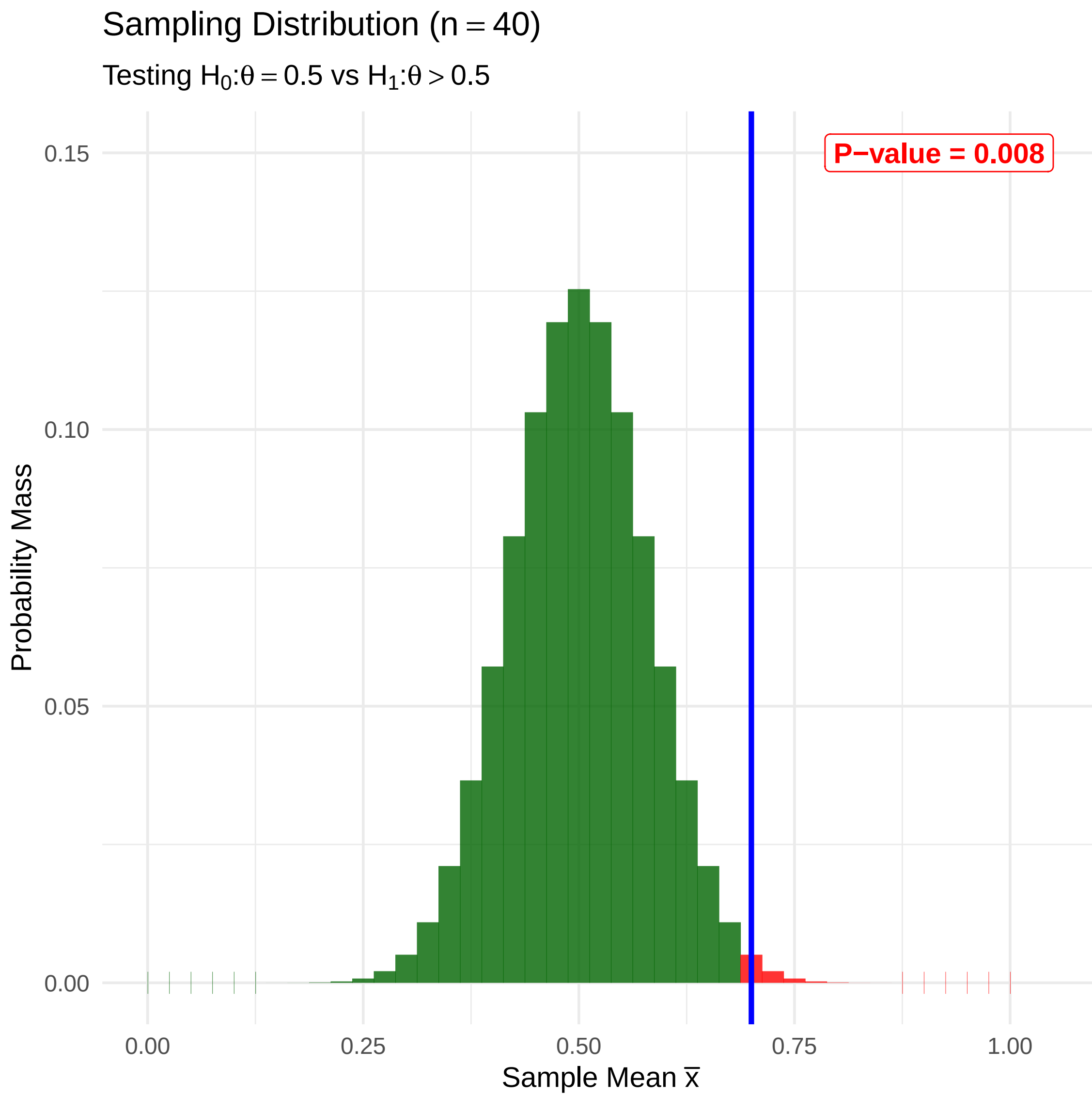

Figure 1.5: Sampling Distribution (n= 40)

1.6.3 Questions to Answer

In this course, we will answer several challenging questions related to general parametric models in the Frequentist framework.

- **MLE:** Can we use the Maximum Likelihood Estimator (MLE) $\hat{\theta}$ for general models even no closed-form solution exists? Is MLE a good method?
- **Sampling Distributions:** What is the distribution of $\hat{\theta}_{\text{MLE}}$? What's its mean and standard deviation?
- **Confidence Intervals:** How to construct CI with $\hat{\theta}$?
- **Hypothesis Testing:** How do we derive powerful tests from the likelihood function? How to assess goodness-of-fit of parametric models with their likelhiood information?

1.7 Bayesian Inference

- **Concept:** θ is regarded as a random variable.
- **Posterior:** Posterior $\propto$ Likelihood $\times$ Prior.

Example: Bayesian Analysis of the Lady Tasting Tea

Prior: Beta$(1, 1)$ (Uniform).

1.7.1 n=10 (k=7)

Posterior: Beta$(1 + 7, 1 + 3) =$ Beta$(8, 4)$

1.7.2 n=40 (k=28)

Posterior: Beta$(1 + 28, 1 + 12) =$ Beta$(29, 13)$.

1.7.3 Questions to Answer

We will also tackle the specific technical challenges involved in Bayesian analysis.

- **Posterior Derivation:** How do we derive the posterior distribution $f(\theta|x)$ for various likelihoods and priors?
- **Comparing with Other methods:** Are Bayesain methods good or not or general inference?
- **Computation:** When the posterior cannot be derived analytically, how do we use computational techniques like Markov Chain Monte Carlo (MCMC) to sample from it?
- **Summarization:** How do we construct Credible Intervals (e.g., Highest Posterior Density regions) from posterior samples?
- **Prediction:** How do we solve the integral required to compute the posterior predictive distribution for future data?
- **Prior:** How to choose our prior? What's its effect on our inference?

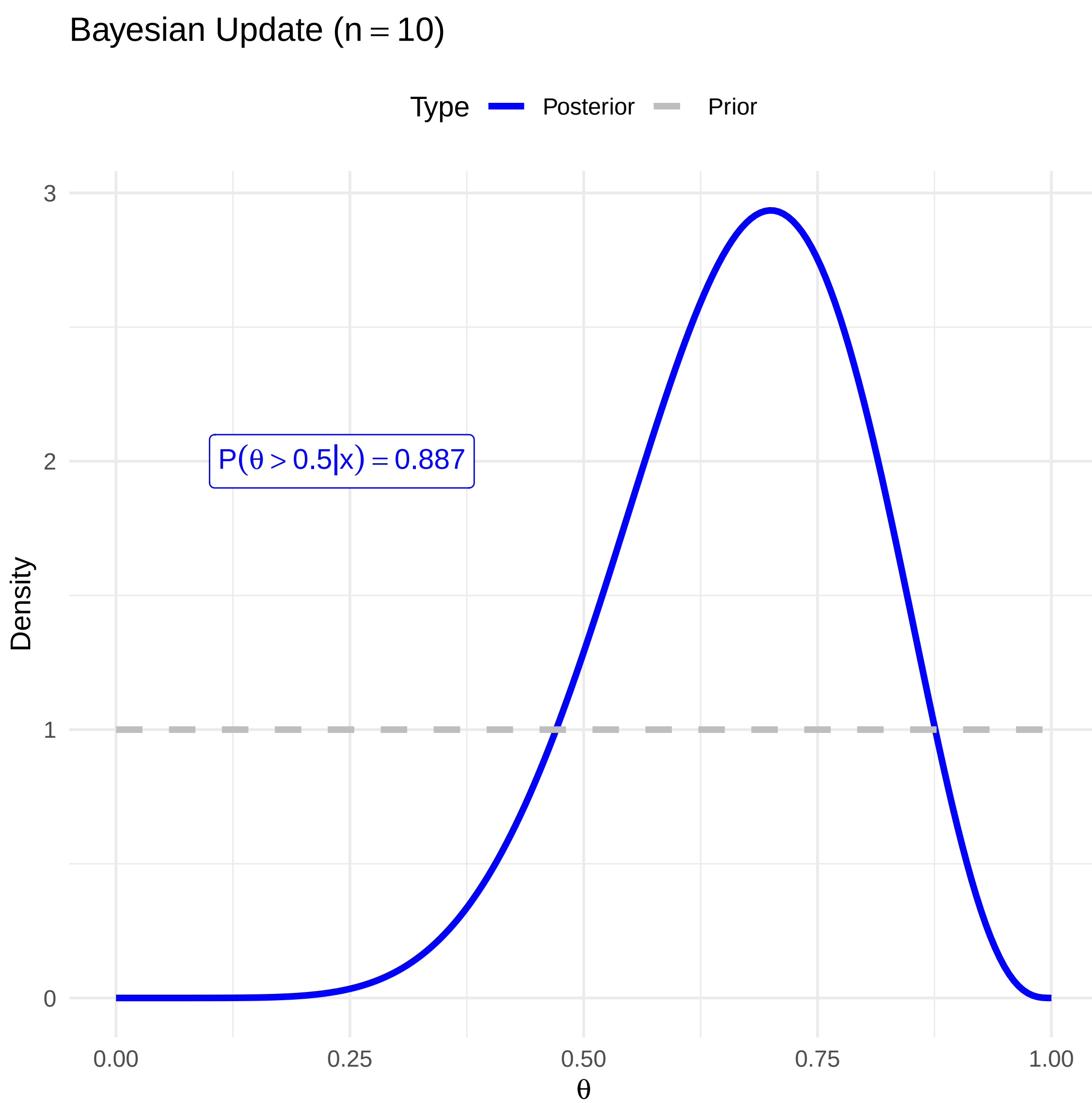

Figure 1.6: Bayesian Update (n= 10)

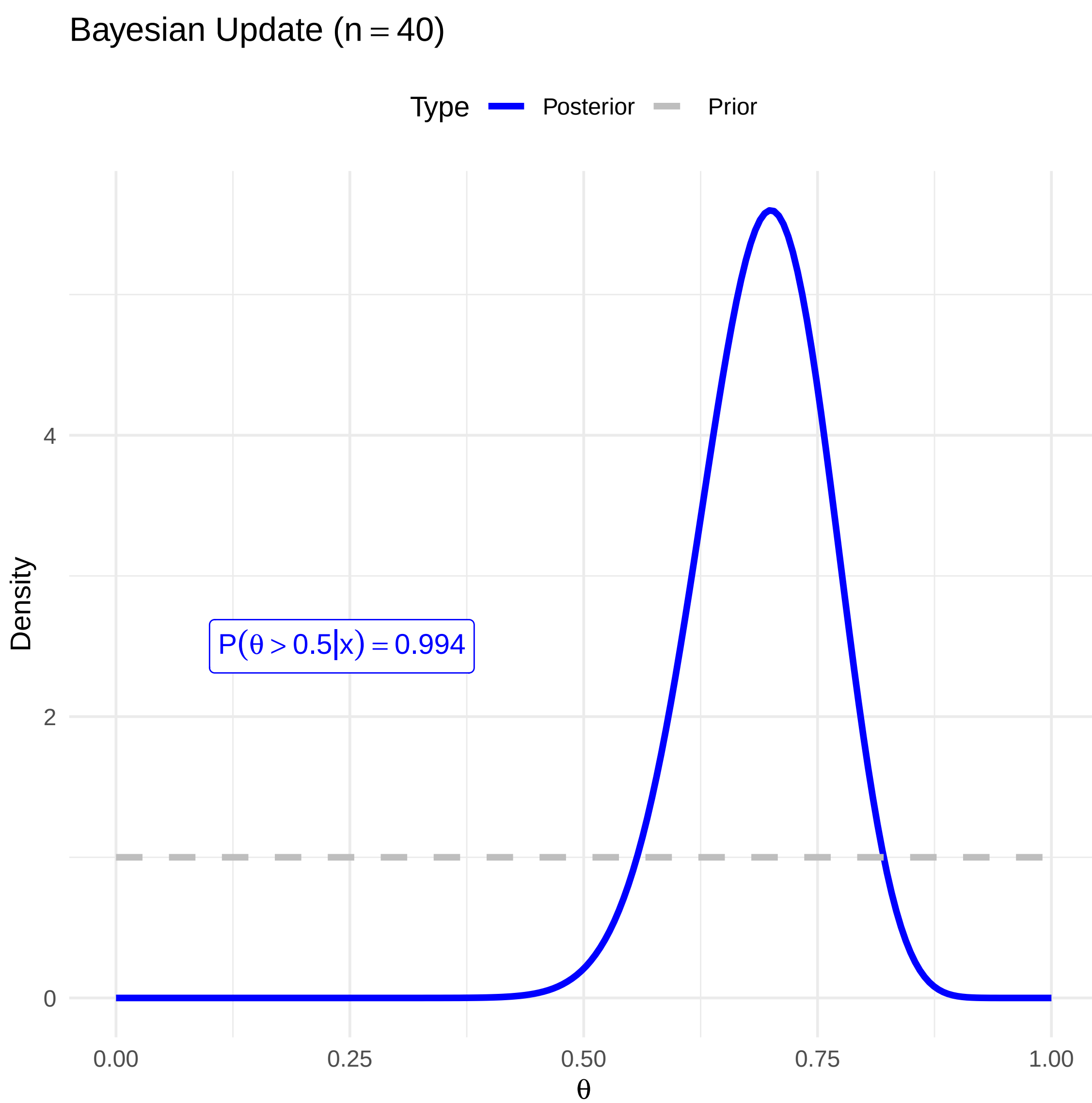

Figure 1.7: Bayesian Update (n= 40)

- **Model Comparison and Assessment:** How to assess a Bayesian model?

2 Introduction to R